# Predictive Modeling

PGP-DSBA Online

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**Context: Problem 1:**

**Linear Regression**:

You are a part of an investment firm and your work is to do research about these 759 firms. You are provided with the dataset containing the sales and other attributes of these 759 firms. Predict the sales of these firms on the bases of the details given in the dataset so as to help your company in investing consciously. Also, provide them with 5 attributes that are most important.

**Data Dictionary for Firm_level_data:**

- **sales**: Sales (in millions of dollars).
- **capital**: Net stock of property, plant, and equipment.
- **patents**: Granted patents.
- **randd**: R&D stock (in millions of dollars).
- **employment**: Employment (in 1000s).
- **sp500**: Membership of firms in the S&P 500 index. S&P is a stock market index that measures the stock performance of 500 large companies listed on stock exchanges in the United States.
- **tobinq**: Tobin's q (also known as q ratio and Kaldor's v) is the ratio between a physical asset's market value and its replacement value.
- **value**: Stock market value.
- **institutions**: Proportion of stock owned by institutions.

**Data Analysis & A detailed solution around the problems:**

**Overview of the dataset:** The initial steps to get an overview of any dataset are to:

- Observe the first few rows of the dataset, to check whether the dataset has been loaded properly or not.
- Get information about the number of rows and columns in the dataset.
- Find out the data types of the columns to ensure that data is stored in the preferred format and the value of each property is as expected.
- Check the statistical summary of the dataset to get an overview of the numerical columns of the data.

**Sample of the dataset:**

| | Unnamed: 0 | sales | capital | patents | randd | employment | sp500 | tobinq | value | institutions |
|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 0 | 826.99505 | 161.60399 | 10 | 382.07825 | 2.30600 | no | 11.04951 | 1625.45376 | 80.27000 |
| **1** | 1 | 407.75397 | 122.10101 | 2 | 0.00000 | 1.86000 | no | 0.84419 | 243.11708 | 59.02000 |
| **2** | 2 | 8407.84559 | 6221.14461 | 138 | 3296.70044 | 49.65900 | yes | 5.20526 | 25865.23380 | 47.70000 |
| **3** | 3 | 451.00001 | 266.89999 | 1 | 83.54016 | 3.07100 | no | 0.30522 | 63.02463 | 26.88000 |
| **4** | 4 | 174.92798 | 140.12400 | 2 | 14.23364 | 1.94700 | no | 1.06330 | 67.40641 | 49.46000 |

**Observation 1:** The dataset has 759 rows and 10 columns.

**Observation 2:** All the columns in the data are non-null, except for column 'tobinq' with missing or null values, which will require treating. Only sp500 column is categorical (object) type rest are numerical (integer) hence it would be easy to analyze the dataset.

**Observation 3:** It's very crucial to identify the missing values, and treat them before moving ahead with the data analysis, and as checked there are 21 missing values in the data, and we will treat all of them before model building.

**Observation 4:** There are 759 unique firms in the data. And a column 'unnamed: 0' with no important data needs to be dropped for further analysis. There are no duplicate values in the data.

**Observation 5:** Statistical summary is nothing but the technique to identify variables such as min, max, mean, standard deviation, count, 25th, 50th, & 75th percentile of the data.

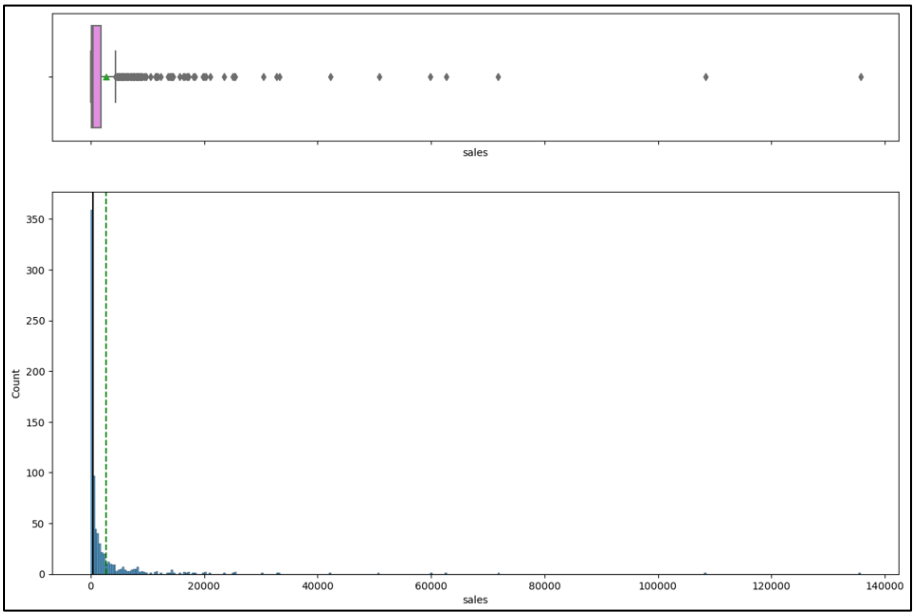
| | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| **sales** | 759.00000 | 2689.70516 | 8722.06012 | 0.13800 | 122.92000 | 448.57708 | 1822.54737 | 135696.78820 |
| **capital** | 759.00000 | 1977.74750 | 6466.70490 | 0.05700 | 52.65050 | 202.17902 | 1075.79002 | 93625.20056 |
| **patents** | 759.00000 | 25.83136 | 97.25958 | 0.00000 | 1.00000 | 3.00000 | 11.50000 | 1220.00000 |
| **randd** | 759.00000 | 439.93807 | 2007.39759 | 0.00000 | 4.62826 | 36.86414 | 143.25340 | 30425.25586 |
| **employment** | 759.00000 | 14.16452 | 43.32144 | 0.00600 | 0.92750 | 2.92400 | 10.05000 | 710.79993 |
| **tobinq** | 738.00000 | 2.79491 | 3.36659 | 0.11900 | 1.01878 | 1.68030 | 3.13931 | 20.00000 |
| **value** | 759.00000 | 2732.73475 | 7071.07236 | 1.97105 | 103.59395 | 410.79353 | 2054.16039 | 95191.59116 |
| **institutions** | 759.00000 | 43.02054 | 21.68559 | 0.00000 | 25.39500 | 44.11000 | 60.51000 | 90.15000 |

**Topic 1: Exploratory Data Analysis:**

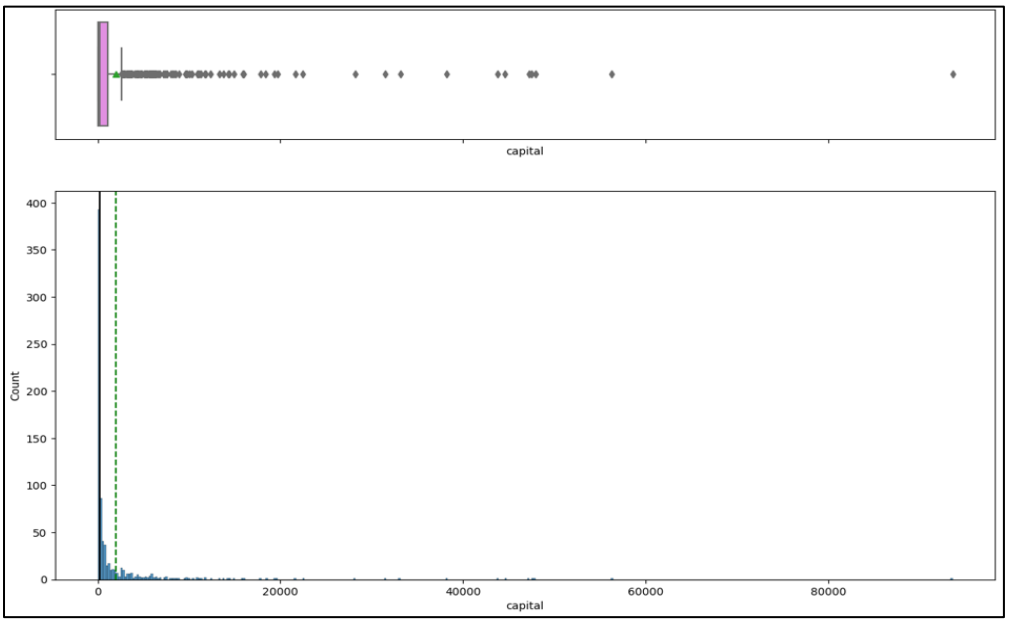
**Observation:** Firms part of sp500 membership are 217 and those not part are 542.

**1. Univariate analysis: (The green dotted lines in all the image denotes average of the data)**

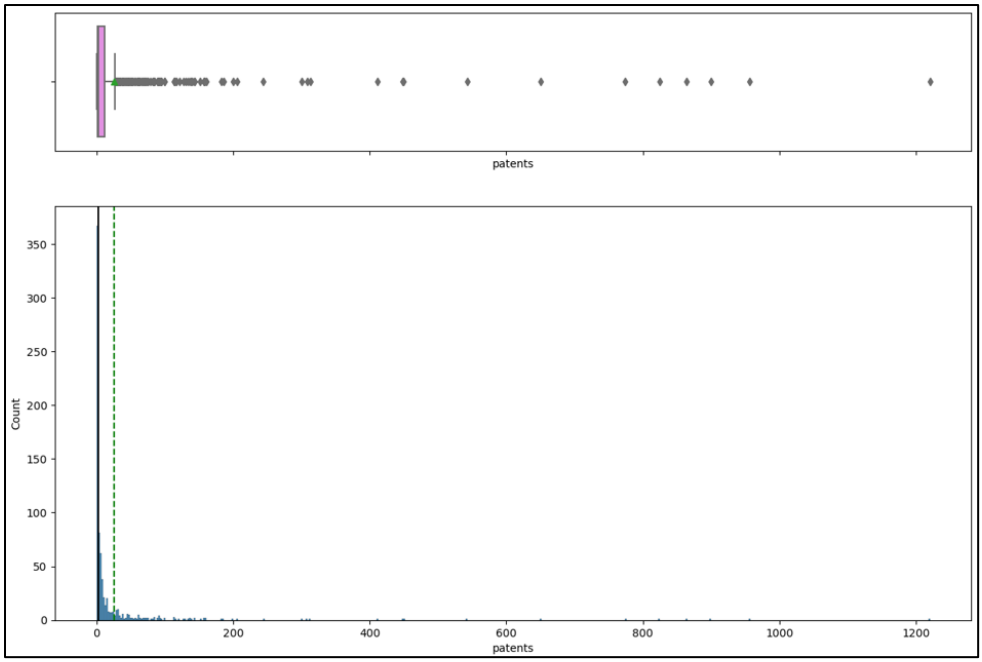
**1.1 Sales:** We see data with lot of outliers, 80% of data ranges between 0.1 to 2600, with an average of 2689 (in million dollars).



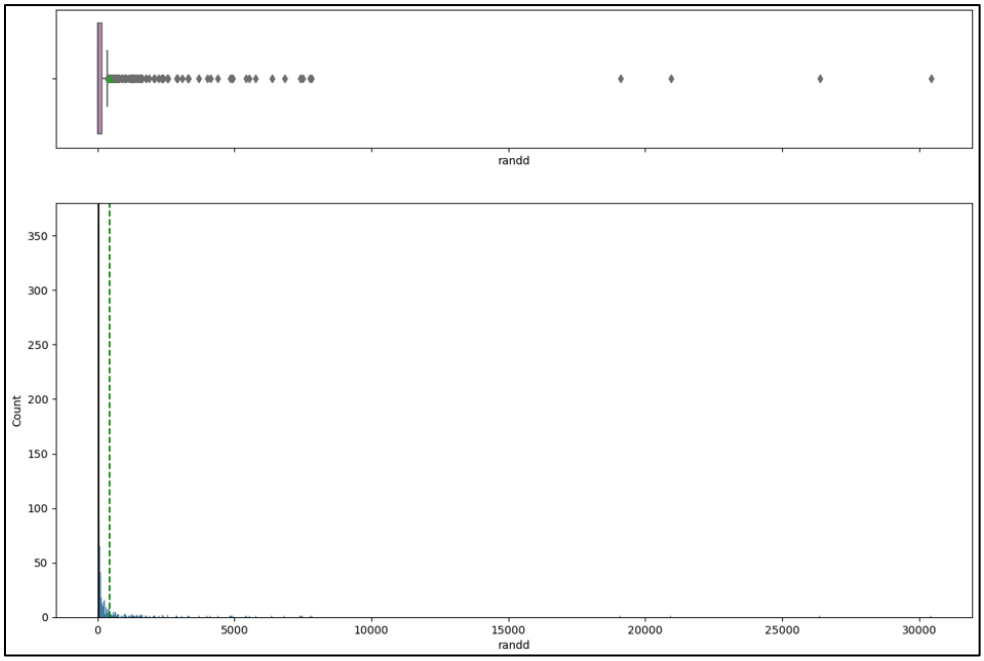

**1.2 Capital:** We see quite many outliers and less uniformity in the data, 80% data lies between 0 to 1900, with an average of 1977.



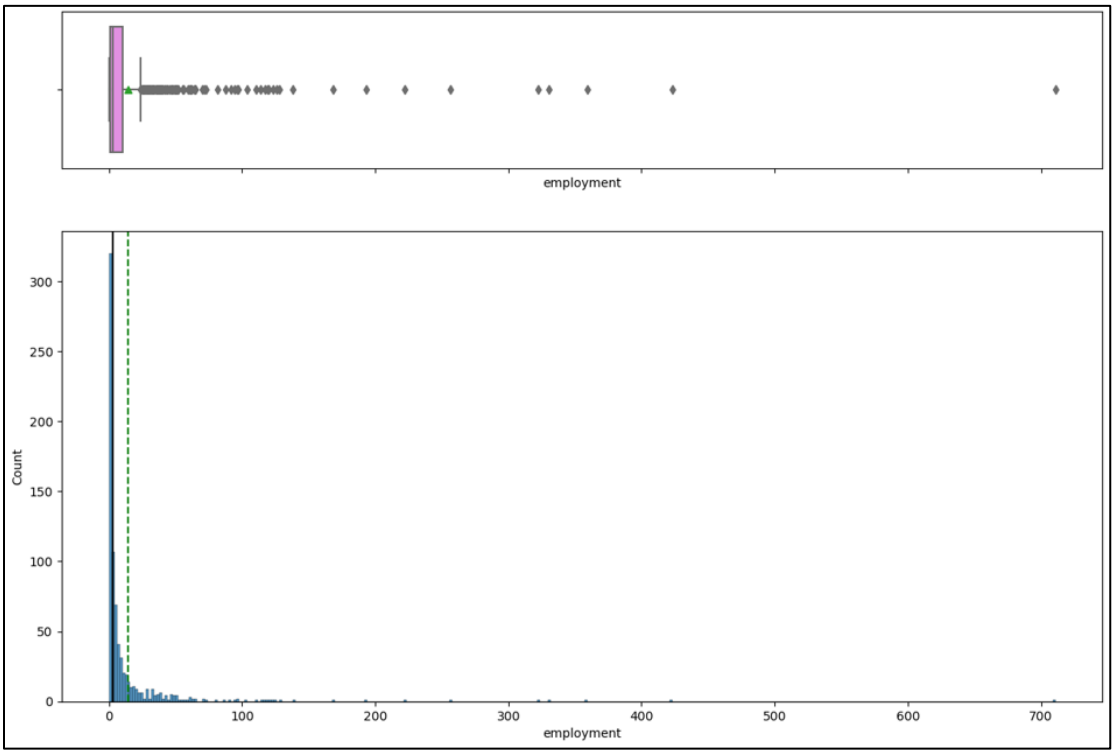
**1.3 Patents:** We observe lot of outliers here too with an average of 25 and 84% data ranging from 0 to 25.



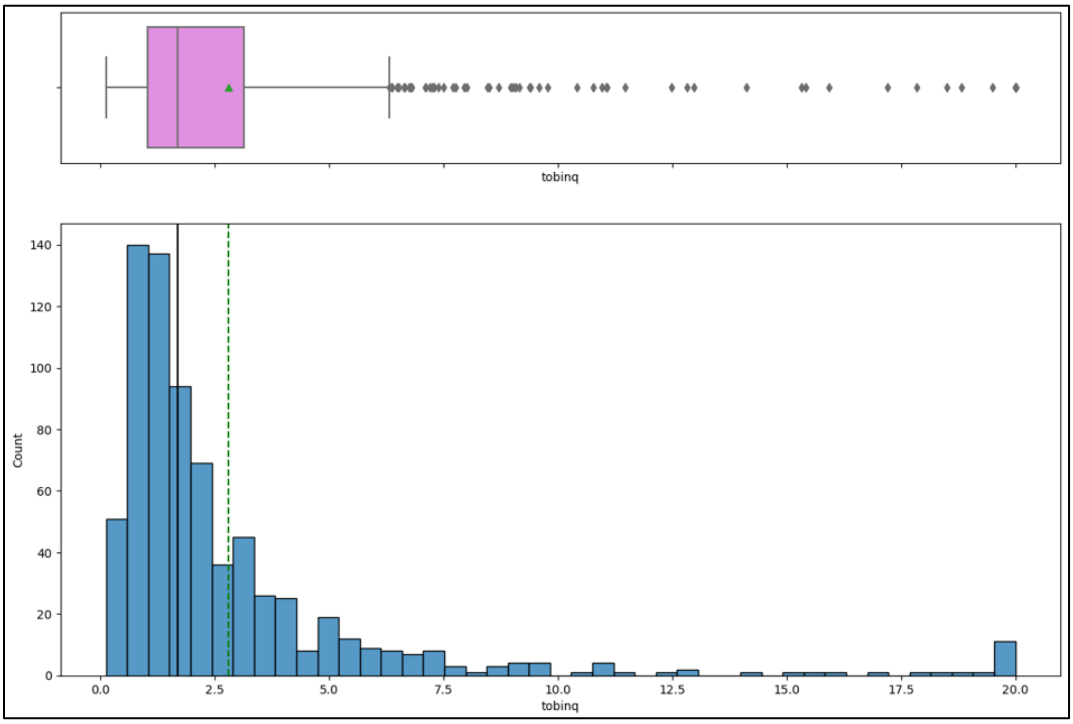

**1.4 R&D (randd):** We see an average of 439, with maximum data ranging from 0 to 400 (in million dollars). There are outliers observed.



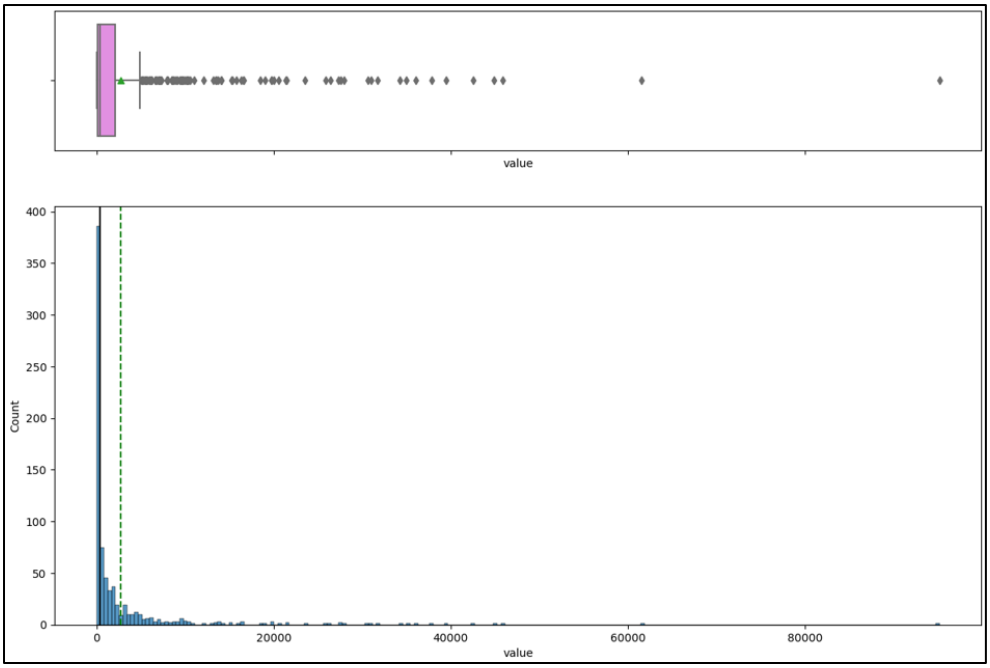
**1.5 Employment:** Average of 14 employees (in 1000s) observed in the data, with many outliers.



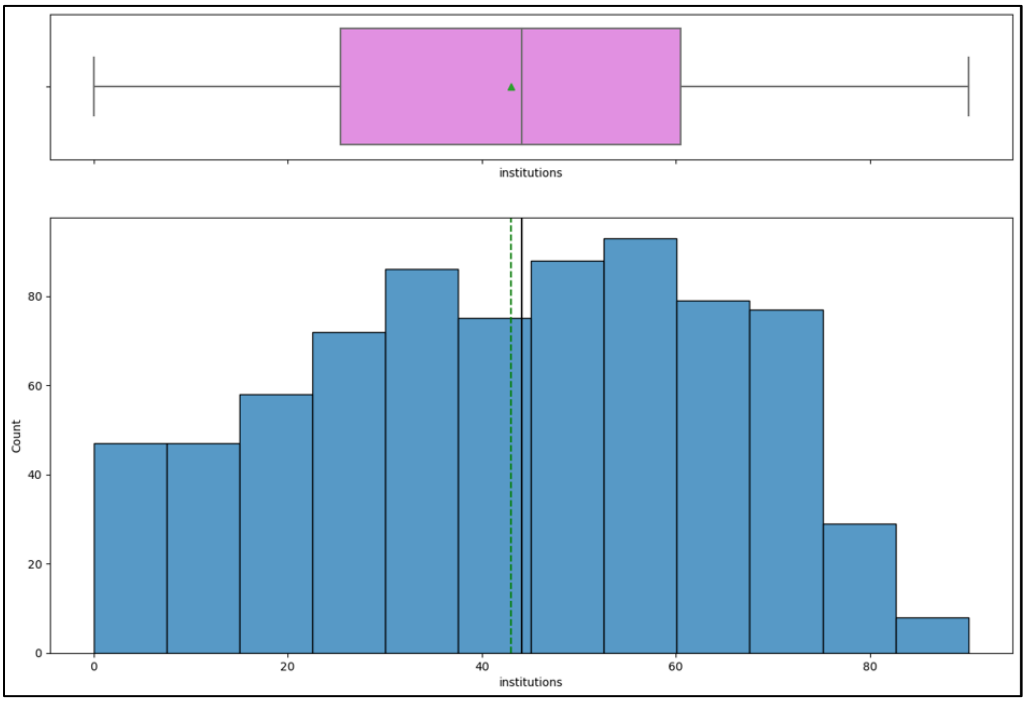

**1.6 Tobinq:** Lot of outliers observed with missing values in the data. The ratio averages at 2.79.



**1.7 Value:** The stock value averages at 2732. With maximum observed between 0 to 2700. Outliers witnessed.

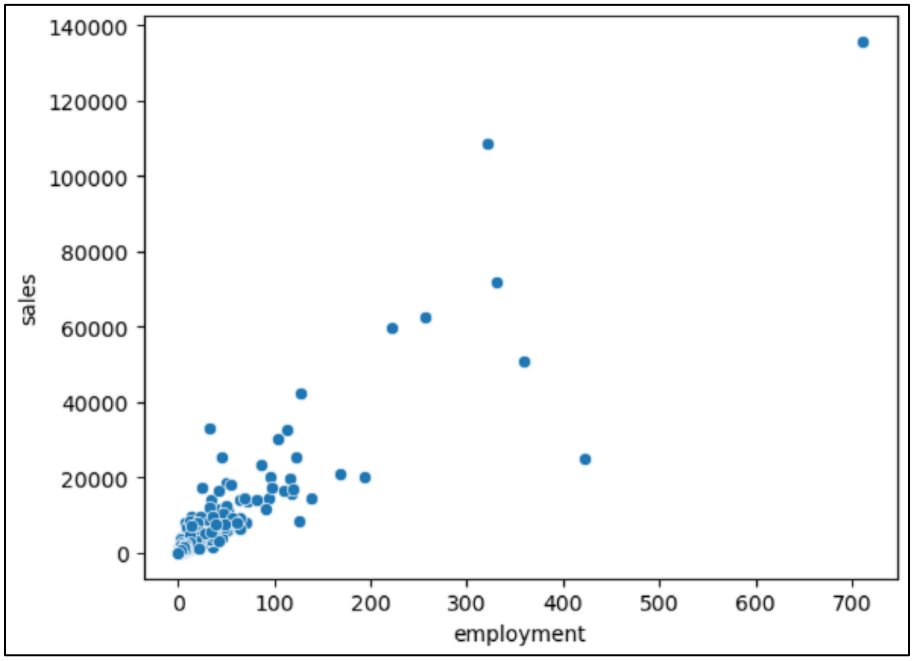



**1.8 Institutions:** Average of 43 stocks owned by institutions. Data quite evenly spread with no outliers.

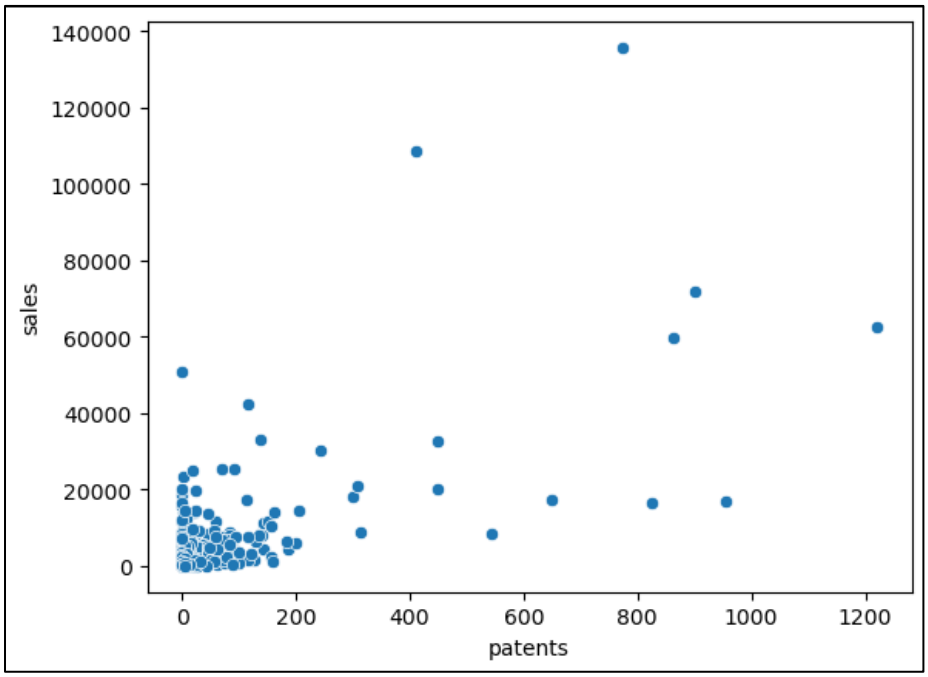


**2. Bivariate Analysis: Let's check for correlations:**

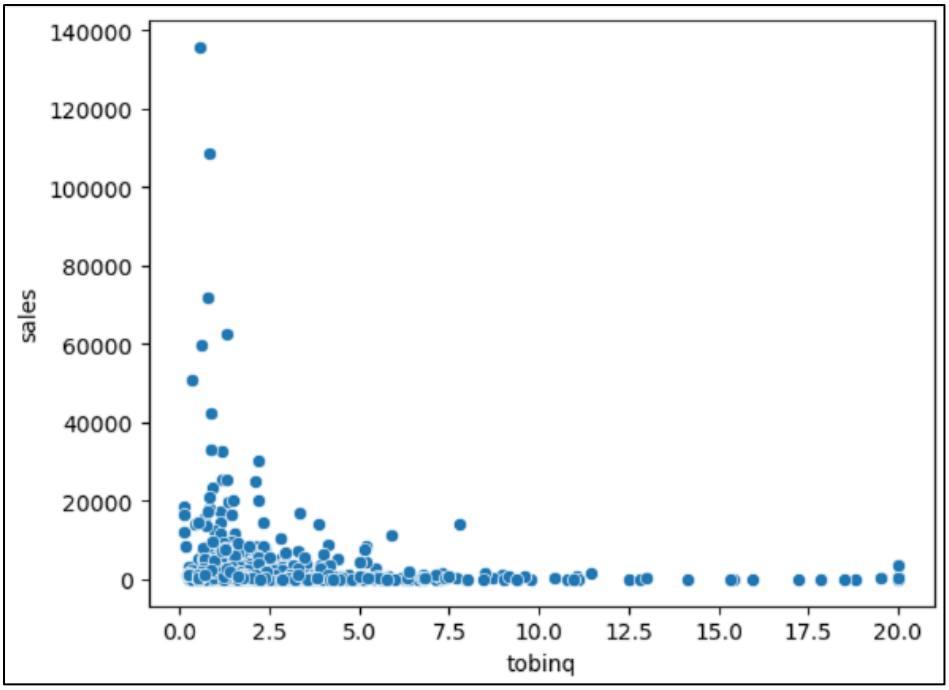
**2.1 Observation of employment on sales:** We observe cluttered pattern between employees and sales. 0 to 200 employees have contributed for sales up to 40000.



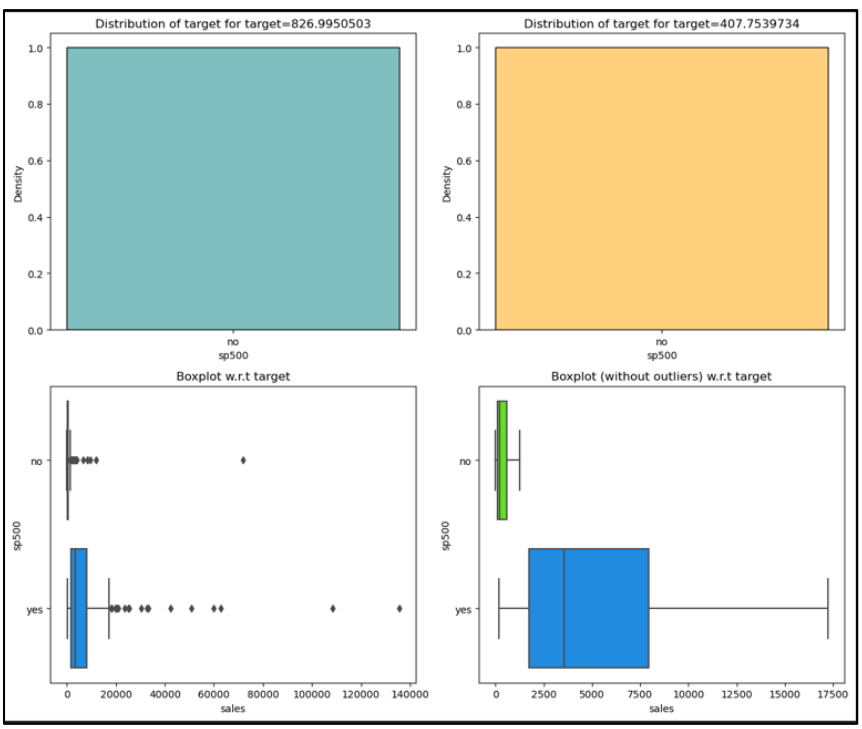

**2.2 Observation of patents on sales:** Uneven spread between patents and sales with a maximum clutter around 0 to 200 patents with close to 22000 of sales.



**2.3 Observation of tobinq on sales:** The scatter quite spread from point one to point 20. With highest sales between 1 to 2.5 with an amount of 140000.

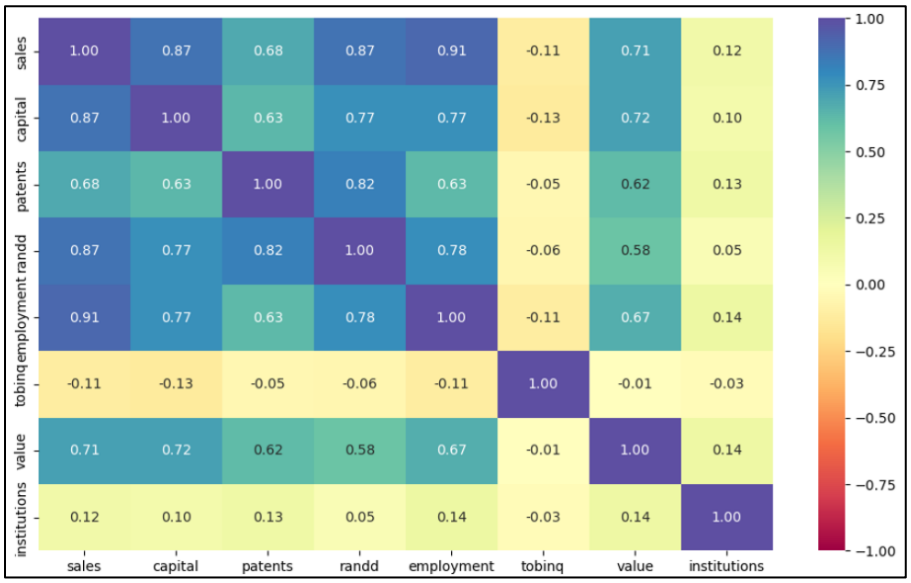



**2.4 Distribution of sales on sp500:** We observe outliers when the target was 827 with maximum sales under 10000 and we observe no outliers when the target was 408 with sales ranging from 2000 to 8000.

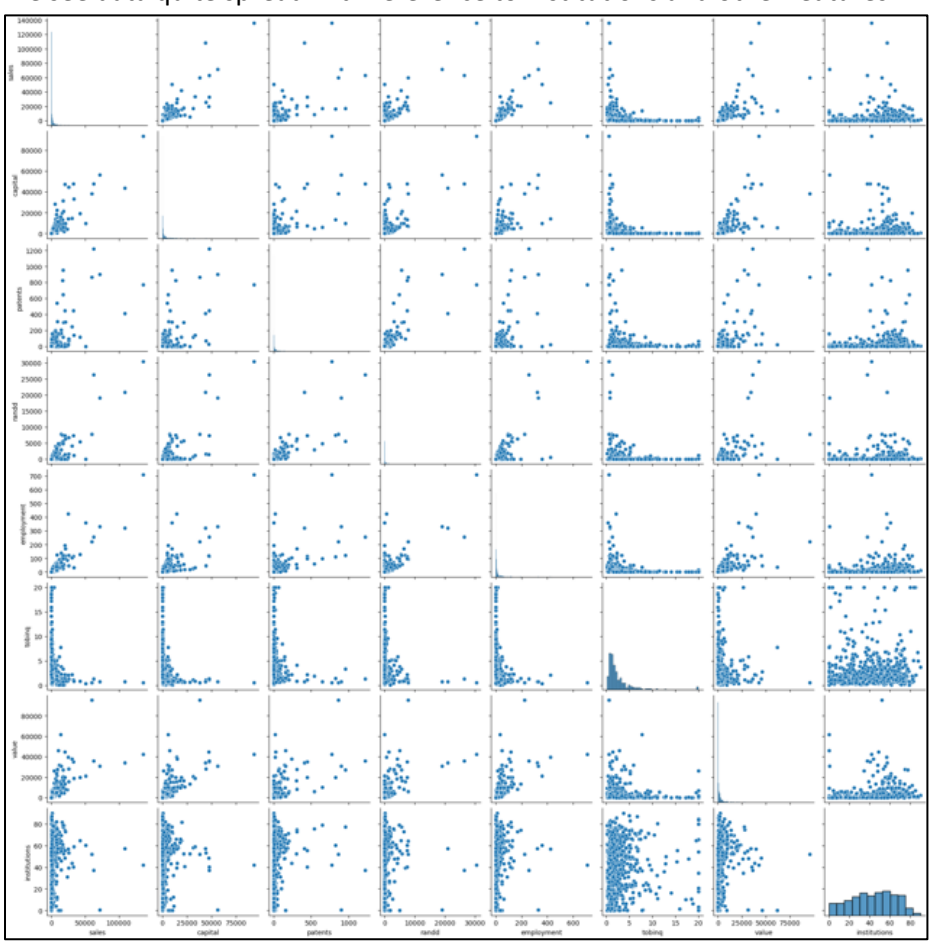


### 2.5 Correlation Map:

- **Heatmap:** We see a positive correlation between sales and employment, followed by sales and RnD along with sales & capital. Quite negative correlation between tobinq and other features.




- **Pairplot:** We see data quite spread with reference to institutions and other features.



## Topic 2: Data Pre-processing:

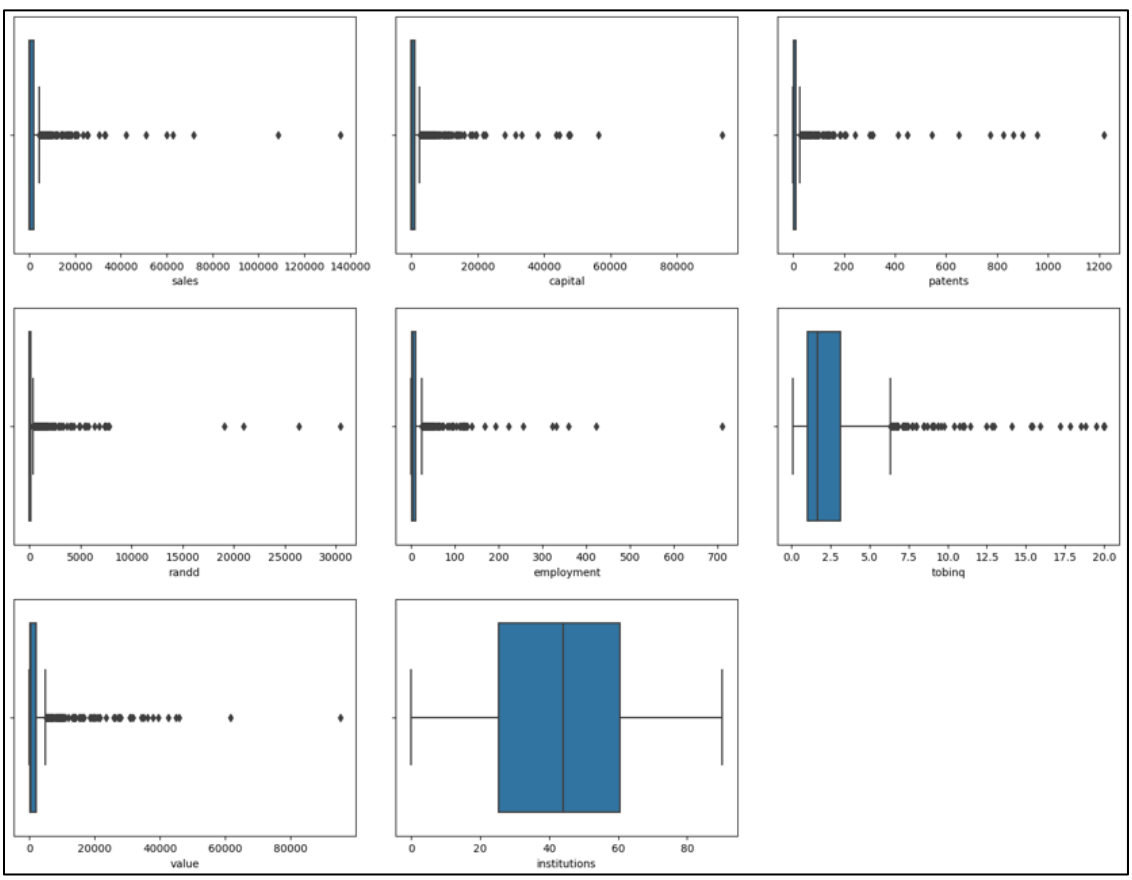
1. **Identifying missing value & treating the same:** 21 missing values found under column 'tobinq'.

To rectify the same we impute the null values with suitable measure and hence fill it with either mean or median values.

| | |
|---|---|
| sales | 0 |
| capital | 0 |
| patents | 0 |
| randd | 0 |
| employment | 0 |
| sp500 | 0 |
| tobinq | 21 |
| value | 0 |
| institutions | 0 |

| | |
|---|---|
| sales | 0 |
| capital | 0 |
| patents | 0 |
| randd | 0 |
| employment | 0 |
| sp500 | 0 |
| tobinq | 0 |
| value | 0 |
| institutions | 0 |

2. **Outlier Detection:** Quite a lot of outliers found in each feature except for institutions.



3. **Encoding:** Defining dependent and independent variables.

```
     capital  patents      randd  employment sp500   tobinq       value
0  161.60399       10  382.07825     2.30600    no 11.04951  1625.45376
1  122.10101        2    0.00000     1.86000    no  0.84419   243.11708
2 6221.14461      138 3296.70044    49.65900   yes  5.20526 25865.23380
3  266.89999        1   83.54016     3.07100    no  0.30522    63.02463
4  140.12400        2   14.23364     1.94700    no  1.06330    67.40641

   institutions
0      80.27000
1      59.02000
2      47.70000
3      26.88000
4      49.46000

0    826.99505
1    407.75397
2   8407.84559
3    451.00001
4    174.92798
```

4. **Creating dummy variables:** Perform dummy encoding on categorical features.

| | const | capital | patents | randd | employment | tobinq | value | institutions | sp500_yes |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1.00000 | 161.60399 | 10 | 382.07825 | 2.30600 | 11.04951 | 1625.45376 | 80.27000 | 0 |
| **1** | 1.00000 | 122.10101 | 2 | 0.00000 | 1.86000 | 0.84419 | 243.11708 | 59.02000 | 0 |
| **2** | 1.00000 | 6221.14461 | 138 | 3296.70044 | 49.65900 | 5.20526 | 25865.23380 | 47.70000 | 1 |
| **3** | 1.00000 | 266.89999 | 1 | 83.54016 | 3.07100 | 0.30522 | 63.02463 | 26.88000 | 0 |
| **4** | 1.00000 | 140.12400 | 2 | 14.23364 | 1.94700 | 1.06330 | 67.40641 | 49.46000 | 0 |

5. **Splitting the data in 70:30 ratios for train to test data.**

- Number of rows in train data = 531
- Number of rows in test data = 228

| | const | capital | patents | randd | employment | tobinq | value | institutions | sp500_yes |
|---|---|---|---|---|---|---|---|---|---|
| **517** | 1.00000 | 87.51000 | 4 | 93.67892 | 1.56700 | 4.92243 | 426.92982 | 49.31000 | 0 |
| **83** | 1.00000 | 939.39882 | 29 | 251.91893 | 15.90000 | 2.33934 | 3221.51235 | 69.97000 | 1 |
| **61** | 1.00000 | 134.42202 | 21 | 114.85410 | 4.50900 | 1.02718 | 499.46127 | 37.68000 | 0 |
| **676** | 1.00000 | 870.27879 | 0 | 0.20930 | 2.70000 | 0.69570 | 432.21225 | 44.11000 | 0 |
| **746** | 1.00000 | 90.60802 | 1 | 0.00000 | 5.08800 | 2.02587 | 220.91163 | 40.94000 | 0 |

## Topic 3: Model Building:

**Step 1:** Train the models with **dependent and independent sets**:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  sales   R-squared:                       0.939
Model:                            OLS   Adj. R-squared:                  0.938
Method:                 Least Squares   F-statistic:                     1009.
Date:                Wed, 20 Mar 2024   Prob (F-statistic):          8.05e-312
Time:                        13:44:19   Log-Likelihood:                -4872.7
No. Observations:                 531   AIC:                             9763.
Df Residuals:                     522   BIC:                             9802.
Df Model:                           8
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const            -93.9839    257.645     -0.365      0.715    -600.133     412.165
capital            0.2534      0.029      8.663      0.000       0.196       0.311
patents          -13.5564      2.078     -6.525      0.000     -17.638      -9.475
randd              1.6949      0.131     12.923      0.000       1.437       1.953
employment       108.6936      5.107     21.284      0.000      98.661     118.726
tobinq           -39.1445     31.810     -1.231      0.219    -101.636      23.347
value              0.1132      0.025      4.569      0.000       0.065       0.162
institutions       4.6284      5.456      0.848      0.397      -6.091      15.347
sp500_yes         50.3527    300.320      0.168      0.867    -539.631     640.336
==============================================================================
Omnibus:                      571.026   Durbin-Watson:                   2.012
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            68046.635
Skew:                           4.577   Prob(JB):                         0.00
Kurtosis:                      57.697   Cond. No.                     2.86e+04
==============================================================================
```

Notes:

- Standard Errors assume that the covariance matrix of the errors is correctly specified.
- The condition number is large, 2.86e+04. This might indicate that there is strong multi-collinearity or other numerical problems.

**Step 2:** Checking model performance on **trained** set:

Training Performance

| | R-squared | Adj. R-squared |
|---|---|---|
| 0 | 0.93925 | 0.93820 |

**Step 3:** Checking model performance on **test** set:

```
Test Performance:
   R-squared  Adj. R-squared
0    0.84826         0.84200
```

**Dealing with high p-value variables:**

- Some of the dummy variables in the data have p-value > 0.05. So, they are not significant and we'll drop them
- But sometimes p-values change after dropping a variable. So, we'll not drop all variables at once
- Instead, we will do the following:
    1. Build a model, check the p-values of the variables, and drop the column with the highest p-value
    2. Create a new model without the dropped feature, check the p-values of the variables, and drop the column with the highest p-value
    3. Repeat the above two steps till there are no columns with p-value > 0.05

**Note**: The above process can also be done manually by picking one variable at a time that has a high p-value, dropping it, and building a model again. But that might be a little tedious and using a loop will be more efficient.

**Step 4:** Now, Building model using **dependent and selected features set**:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  sales   R-squared (uncentered):              0.944
Model:                            OLS   Adj. R-squared (uncentered):         0.943
Method:                 Least Squares   F-statistic:                         1769.
Date:                Thu, 21 Mar 2024   Prob (F-statistic):                   0.00
Time:                        15:22:01   Log-Likelihood:                     -4874.0
No. Observations:                 531   AIC:                                 9758.
Df Residuals:                     526   BIC:                                 9779.
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
capital        0.2583      0.029      8.944      0.000       0.202       0.315
patents      -13.2015      2.044     -6.460      0.000     -17.216      -9.187
randd          1.6530      0.125     13.187      0.000       1.407       1.899
employment   110.2488      4.908     22.462      0.000     100.606     119.891
value          0.1108      0.022      5.003      0.000       0.067       0.154
==============================================================================
Omnibus:                      579.215   Durbin-Watson:                   2.025
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            71668.226
Skew:                           4.681   Prob(JB):                         0.00
Kurtosis:                      59.139   Cond. No.                         458.
==============================================================================
```

Notes:

- $R^2$ is computed without centering (uncentered) since the model does not contain a constant.
- Standard Errors assume that the covariance matrix of the errors is correctly specified.

**Step 5:** Checking model performance on **trained** set:

```
Training Performance
```

| | R-squared | Adj. R-squared |
|---|---|---|
| 0 | 0.93894 | 0.93836 |

**Step 6:** Checking model performance on **test** set:

```
Test Performance:
   R-squared  Adj. R-squared
0    0.84660         0.84315
```

**Actionable Insights:**

- We observe increase in sales with increase in employment.
- Similarly, increase in capital also affects increase in sales, keeping other predictors constant.
- There are also some negative co-efficients, which implies when negative co-efficients are decreased the sales also decreases.

**Business Recommendation:**

- The investment criteria for any new investor are mainly based on the capital invested in the company by the promoters and investors are competing on the firms where the capital investment is good.
- Investment criteria also depends on factor that company has a good sale which is shown by scatter plot.
- To generate capital the company should have the combination of the following attributes such as value, employment, sales, and patents.
- The highest contributing attribute is employment followed by patents.
- Company should provide more employment as increase in sales can only be done by hiring more no of people.
- As sales increases, definitely any new firm will be highly interested to invest.

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**Context: Problem 2:**

**Logistic Regression and Linear Discriminant Analysis:**

You are hired by the Government to do an analysis of car crashes. You are provided details of car crashes, among which some people survived and some didn't. You have to help the government in predicting whether a person will survive or not on the basis of the information given in the data set so as to provide insights that will help the government to make stronger laws for car manufacturers to ensure safety measures. Also, find out the important factors on the basis of which you made your predictions.

**Data Dictionary for Car_Crash:**

1. **dvcat:** factor with levels (estimated impact speeds) 1-9km/h, 10-24, 25-39, 40-54, 55+
2. **weight:** Observation weights, albeit of uncertain accuracy, designed to account for varying sampling probabilities. (The inverse probability weighting estimator can be used to demonstrate causality when the researcher cannot conduct a controlled experiment but has observed data to model).
3. **Survived:** factor with levels Survived or not_survived
4. **airbag:** a factor with levels none or airbag
5. **seatbelt:** a factor with levels none or belted
6. **frontal:** a numeric vector; 0 = non-frontal, 1=frontal impact
7. **sex:** a factor with levels f: Female or m: Male
8. **ageOFocc:** age of occupant in years
9. **yearacc:** year of accident
10. **yearVeh:** Year of model of vehicle; a numeric vector
11. **abcat:** Did one or more (driver or passenger) airbag(s) deploy? This factor has levels deploy, no deploy and unavail.
12. **occRole:** a factor with levels driver or pass: passenger
13. **deploy:** a numeric vector: 0 if an airbag was unavailable or did not deploy; 1 if one or more bags deployed.
14. **injSeverity:** a numeric vector; 0: none, 1: possible injury, 2: no incapacity, 3: incapacity, 4: killed; 5: unknown, 6: prior death
15. **caseid:** character, created by pasting together the populations sampling unit, the case number, and the vehicle number. Within each year, use this to uniquely identify the vehicle.

**Data Analysis & A detailed solution around the problems:**

**Overview of the dataset:** The initial steps to get an overview of any dataset are to:

- Observe the first few rows of the dataset, to check whether the dataset has been loaded properly or not.
- Get information about the number of rows and columns in the dataset.
- Find out the data types of the columns to ensure that data is stored in the preferred format and the value of each property is as expected.
- Check the statistical summary of the dataset to get an overview of the numerical columns of the data.

**Sample of the dataset:**

| | Unnamed: 0 | dvcat | weight | Survived | airbag | seatbelt | frontal | sex | ageOFocc | yearacc | yearVeh | abcat | occRole | deploy | injSeverity | caseid |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 55+ | 27.07800 | Not_Survived | none | none | 1 | m | 32 | 1997 | 1987.00000 | unavail | driver | 0 | 4.00000 | 2:13:2 |
| 1 | 1 | 25-39 | 89.62700 | Not_Survived | airbag | belted | 0 | f | 54 | 1997 | 1994.00000 | nodeploy | driver | 0 | 4.00000 | 2:17:1 |
| 2 | 2 | 55+ | 27.07800 | Not_Survived | none | belted | 1 | m | 67 | 1997 | 1992.00000 | unavail | driver | 0 | 4.00000 | 2:79:1 |
| 3 | 3 | 55+ | 27.07800 | Not_Survived | none | belted | 1 | f | 64 | 1997 | 1992.00000 | unavail | pass | 0 | 4.00000 | 2:79:1 |
| 4 | 4 | 55+ | 13.37400 | Not_Survived | none | none | 1 | m | 23 | 1997 | 1986.00000 | unavail | driver | 0 | 4.00000 | 4:58:1 |

**Observation 1:** The dataset has 11217 rows and 16 columns.

**Observation 2:** All the columns in the data are non-null, except for column 'injSeverity' with missing or null values, which will require treating. We have 8 columns which are categorical (object).

**Observation 3:** It's very crucial to identify the missing values, and treat them before moving ahead with the data analysis, and as checked there are 71 missing values in the data, and we will treat them before model building.

**Observation 4:** There are 11217 unique crashes in the data. And a column 'unnamed: 0' with no important data needs to be dropped for further analysis. There are no duplicate values in the data.

**Observation 5:** Statistical summary is nothing but the technique to identify variables such as min, max, mean, standard deviation, count, 25th, 50th, & 75th percentile of the data.

| | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| Unnamed: 0 | 11217.00000 | 5608.00000 | 3238.21332 | 0.00000 | 2804.00000 | 5608.00000 | 8412.00000 | 11216.00000 |
| weight | 11217.00000 | 431.40531 | 1406.20294 | 0.00000 | 28.29200 | 82.19500 | 324.05600 | 31694.04000 |
| frontal | 11217.00000 | 0.64402 | 0.47883 | 0.00000 | 0.00000 | 1.00000 | 1.00000 | 1.00000 |
| ageOFocc | 11217.00000 | 37.42765 | 18.19243 | 16.00000 | 22.00000 | 33.00000 | 48.00000 | 97.00000 |
| yearacc | 11217.00000 | 2001.10324 | 1.05681 | 1997.00000 | 2001.00000 | 2001.00000 | 2002.00000 | 2002.00000 |
| yearVeh | 11217.00000 | 1994.17794 | 5.65870 | 1953.00000 | 1991.00000 | 1995.00000 | 1999.00000 | 2003.00000 |
| deploy | 11217.00000 | 0.38914 | 0.48758 | 0.00000 | 0.00000 | 0.00000 | 1.00000 | 1.00000 |
| injSeverity | 11140.00000 | 1.82558 | 1.37854 | 0.00000 | 1.00000 | 2.00000 | 3.00000 | 5.00000 |

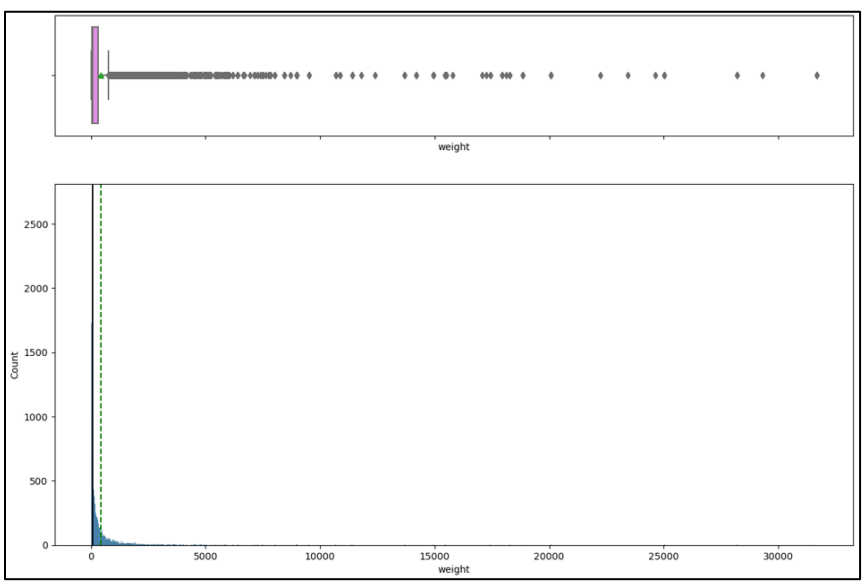
## Topic 1: Exploratory Data Analysis:

Categorical data listed below:

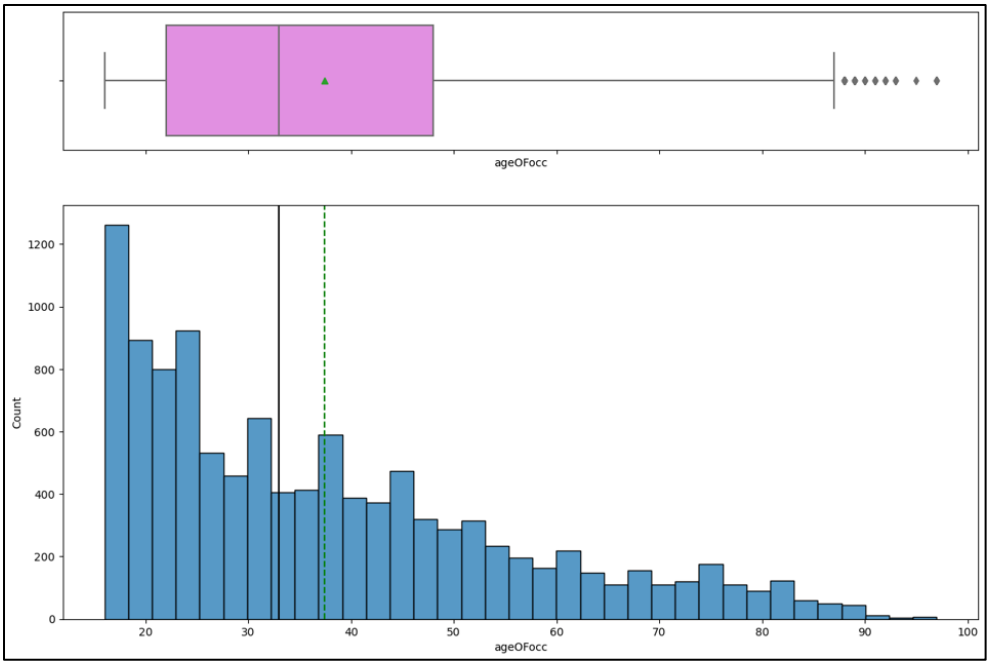
```
10-24      5414
25-39      3368
40-54      1344
55+         809
1-9km/h     282
Name: dvcat, dtype: int64
-----------------------------------------------
survived        10037
Not_Survived     1180
Name: Survived, dtype: int64
-----------------------------------------------
airbag    7064
none      4153
Name: airbag, dtype: int64
-----------------------------------------------
belted    7849
none      3368
Name: seatbelt, dtype: int64
-----------------------------------------------
m    6048
f    5169
Name: sex, dtype: int64
-----------------------------------------------
deploy      4365
unavail     4153
nodeploy    2699
Name: abcat, dtype: int64
-----------------------------------------------
driver    8786
pass      2431
Name: occRole, dtype: int64
```

1. **Univariate Analysis:**

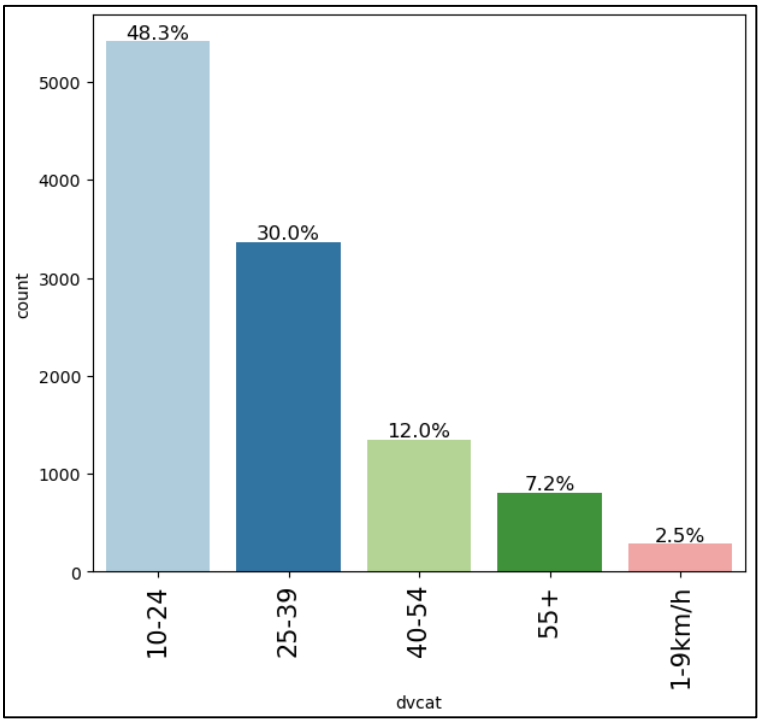
**1.1 Observation on weight:** Lot of outliers observed, with an average of weight of 431.



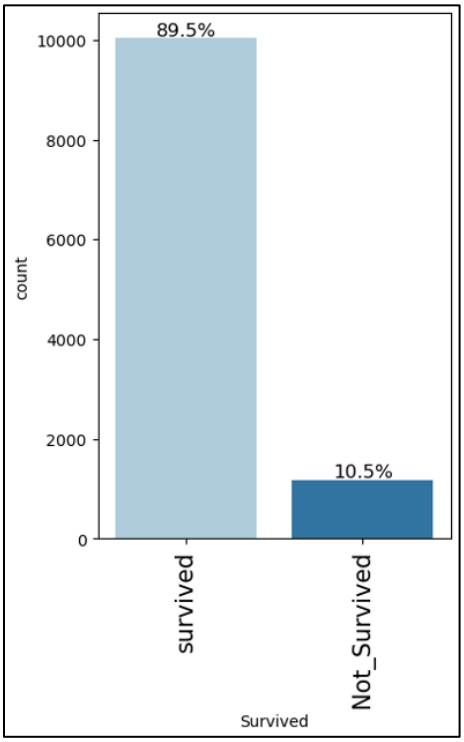
**1.2 Observation on ageOFocc:** Data quite spread with few outliers, average age of 37 observed. We see age of occupants starting from 16 going up to 97. We have 78% data falling under age 50 or less.



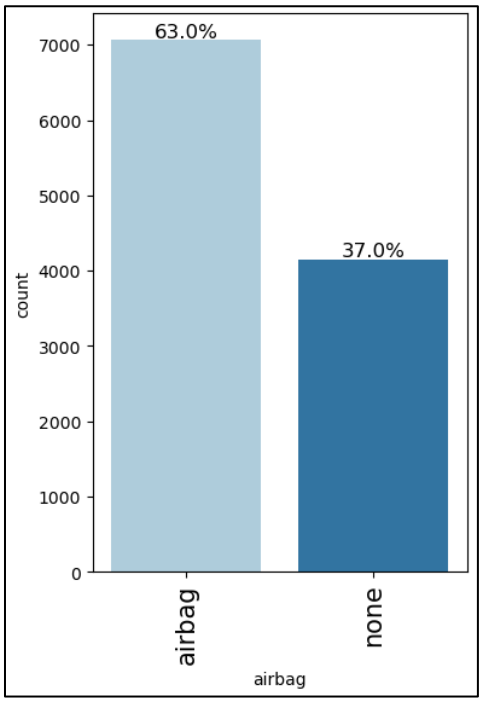

**1.3 Observation on dvcat:** 48% data observed between 10-24kmph speed, followed by 30% between 25-39kmph.



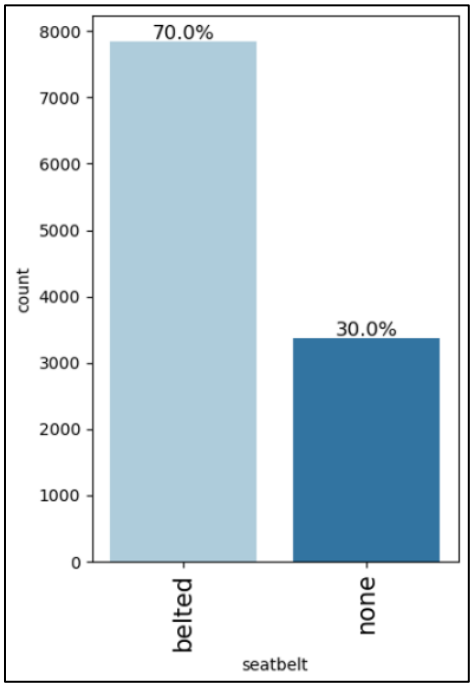
**1.4 Observation on Survived:** 89.5% have survived the crash, with 10.5% life lost (not survived).



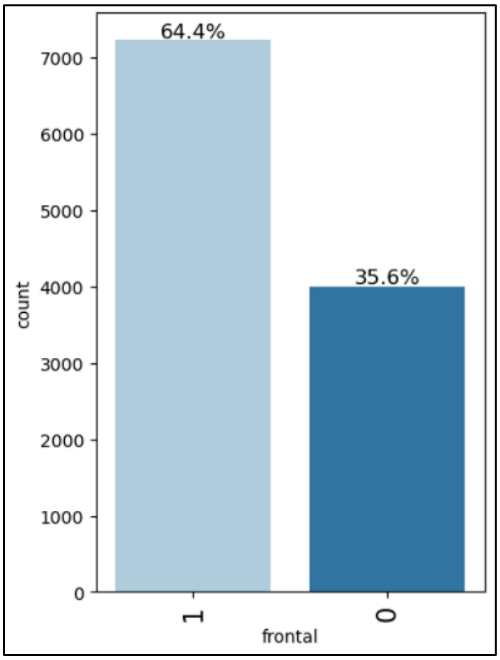

**1.5 Observation on airbag:** During the crash only 63% of them had airbags in the car, with 37% of them without airbags.



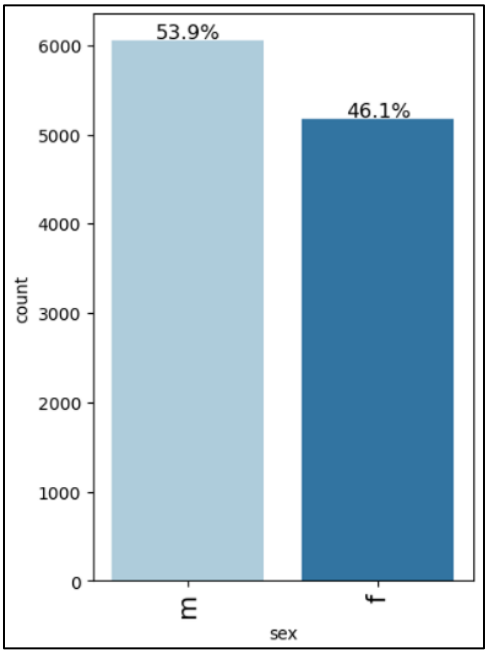
**1.6 Observation on seatbelt:** 70% of them were found with seatbelt on, and 30% without seatbelt on.



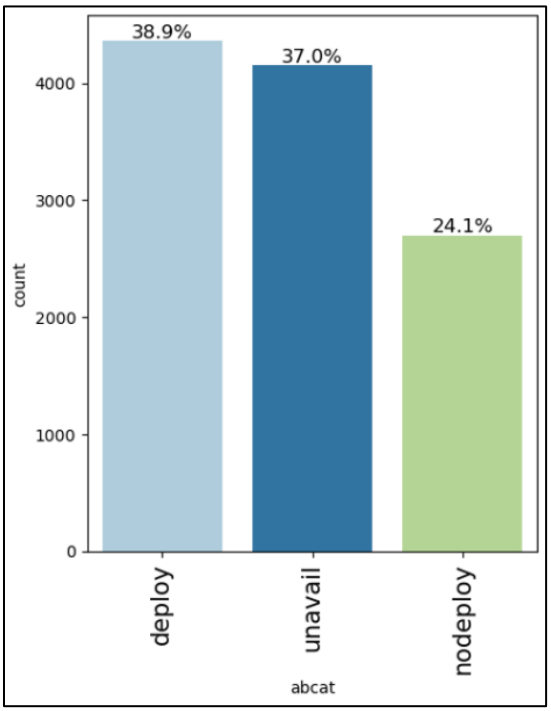

**1.7 Observation on frontal:** The crash had frontal impact to 64% cars, with no frontal impact to 36% cars.



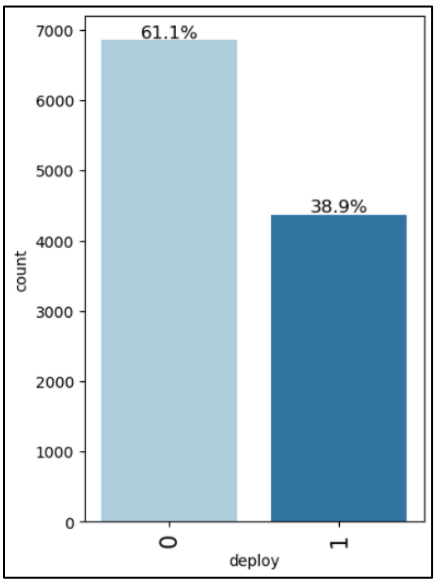
**1.8 Observation on sex:** 54% were male and 46% were female, as observed in the data.



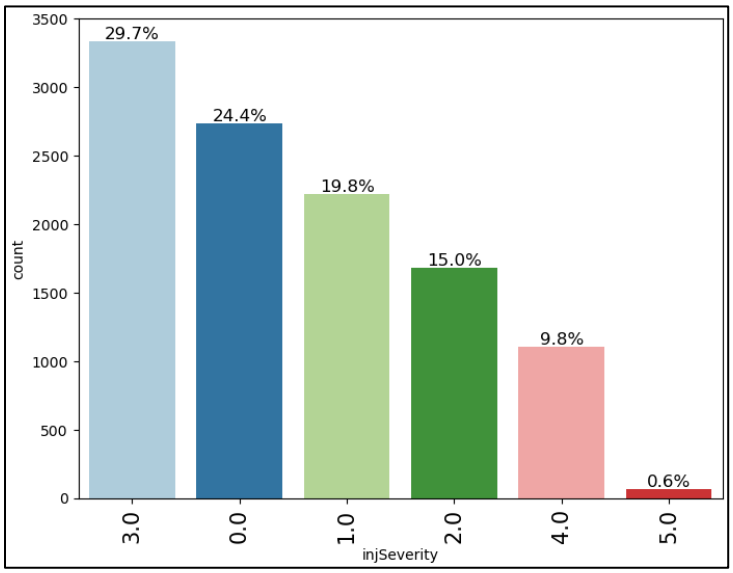

**1.9 Observation on abcat:** Airbags deployed for 39%, was unavailable for the 37% and did not deploy for the rest 24%.



**1.10 Observation on deploy:** Only 39% times the airbags deployed, as compared to airbags either wasn't available or did not deploy with 61%.



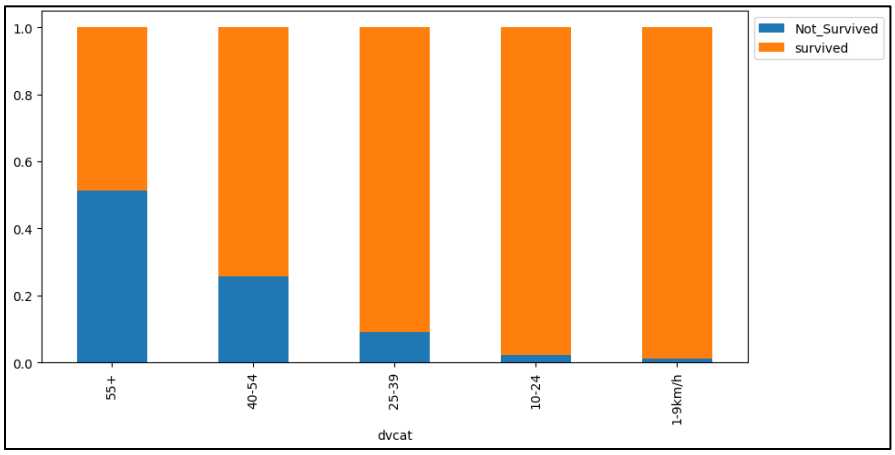

**1.11 Observations on injSeverity:** 24.4% were not injured at all, 19.8% had possible injury, 15% had no disability with the injury, 29.7% were injured with certain disability (Incapacitating injuries are injuries that prevent someone from driving, walking, or performing normal activities, but are not fatal), 9.8% were killed in the crash, and 0.6% were unknown.



## 2. Bivariate Analysis:

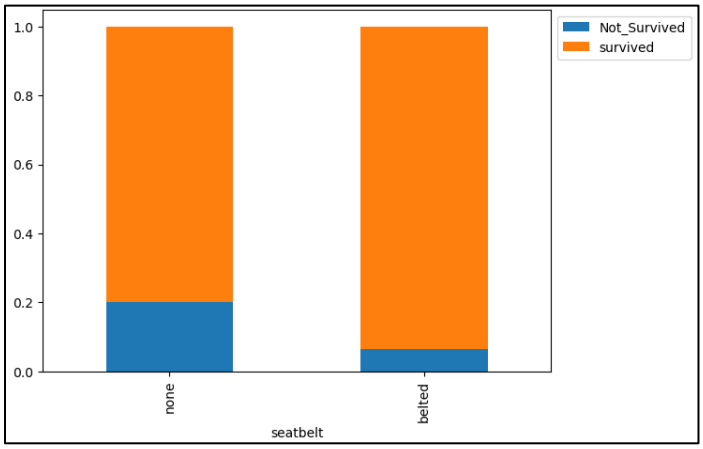
**2.1 Observation on dvcat and Survived:** It's clear that more have survived at a lesser speed, as compared to people with high speed; only 48% survived with speed more than 55+ kmph.

| Survived | Not_Survived | survived | All |
|---|---|---|---|
| dvcat | | | |
| All | 1180 | 10037 | 11217 |
| 55+ | 415 | 394 | 809 |
| 40-54 | 344 | 1000 | 1344 |
| 25-39 | 304 | 3064 | 3368 |
| 10-24 | 114 | 5300 | 5414 |
| 1-9km/h | 3 | 279 | 282 |



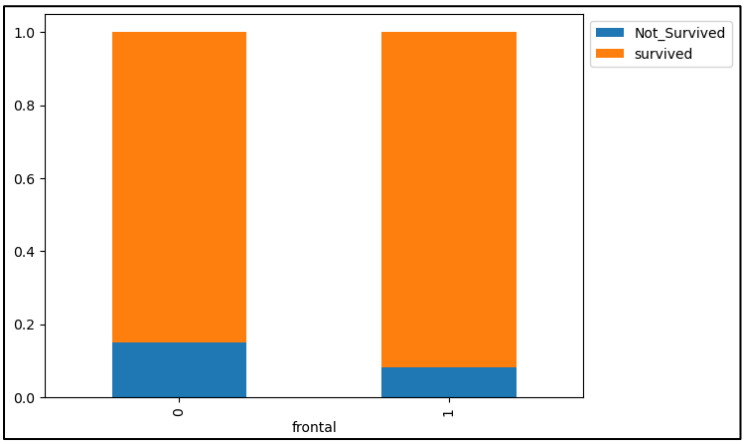

**2.2 Observation on seatbelt and Survived:** 93% survived with seatbelt on, and only 79% survived without seatbelt.

| Survived | Not_Survived | survived | All |
|---|---|---|---|
| seatbelt | | | |
| All | 1180 | 10037 | 11217 |
| none | 680 | 2688 | 3368 |
| belted | 500 | 7349 | 7849 |

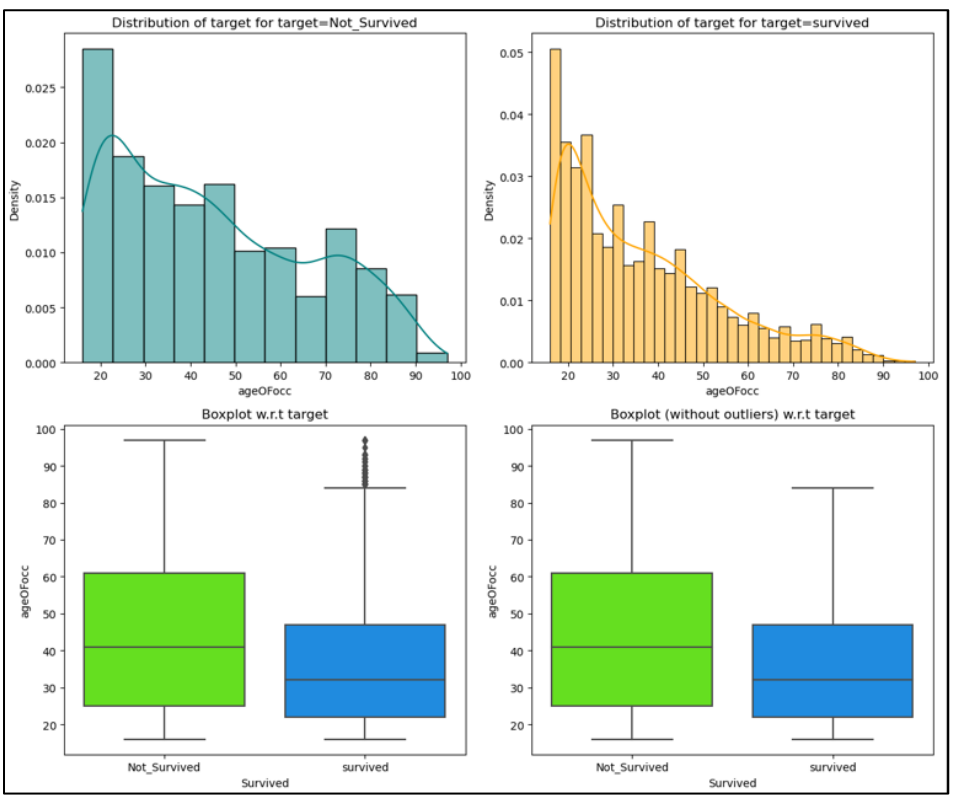


**2.3 Observation on frontal and Survived:** 92% survived despite the frontal impact, with 8% casualty. And only 85% survived with no frontal impact, with 15% casualty.

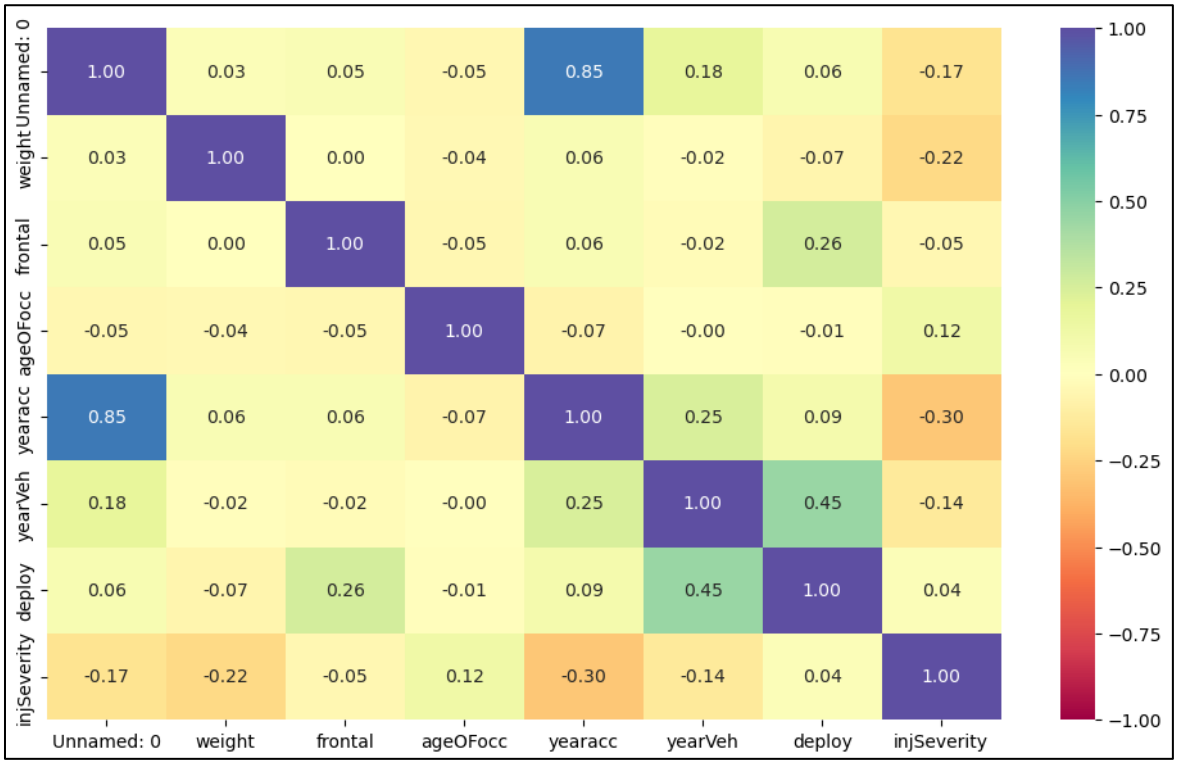
| Survived | Not_Survived | survived | All |
|---|---|---|---|
| frontal | | | |
| All | 1180 | 10037 | 11217 |
| 0 | 598 | 3395 | 3993 |
| 1 | 582 | 6642 | 7224 |




**2.4 Distribution of ageOFocc on Survived:** We see age of occupants starting from 16 going up to 97. We have 78% data falling under age 50 or less. We observe a 56% survival among age group 16 to 35. We also notice a 50% non-survival among age group 16 to 40.



**2.5 Correlation Map:** We witness no correlation between features. In fact quite negative correlation observed.



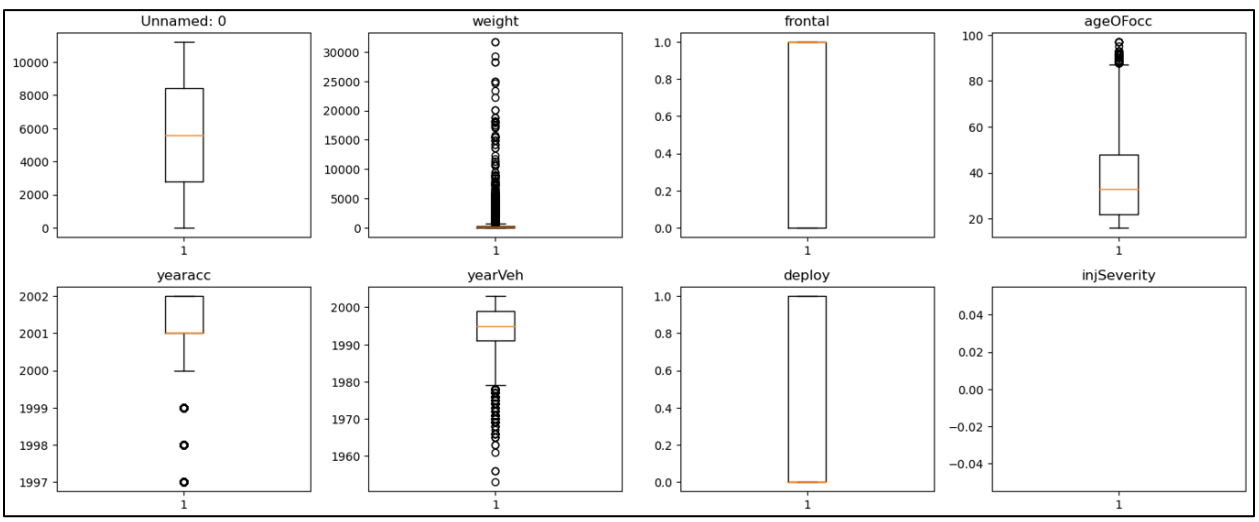
## Topic 2: Data pre-processing:

1. **Missing value treatment:** To rectify the same we impute the null values with suitable measure and hence fill it with either mean or median values.

| | |
|---|---|
| Unnamed: 0 | 0 |
| dvcat | 0 |
| weight | 0 |
| Survived | 0 |
| airbag | 0 |
| seatbelt | 0 |
| frontal | 0 |
| sex | 0 |
| ageOFocc | 0 |
| yearacc | 0 |
| yearVeh | 0 |
| abcat | 0 |
| occRole | 0 |
| deploy | 0 |
| injSeverity | 77 |
| caseid | 0 |

| | |
|---|---|
| Unnamed: 0 | 0 |
| dvcat | 0 |
| weight | 0 |
| Survived | 0 |
| airbag | 0 |
| seatbelt | 0 |
| frontal | 0 |
| sex | 0 |
| ageOFocc | 0 |
| yearacc | 0 |
| yearVeh | 0 |
| abcat | 0 |
| occRole | 0 |
| deploy | 0 |
| injSeverity | 0 |
| caseid | 0 |

2. **Outlier Check:** We can see outliers in few features like weight, age of occupant, year of accident and year of model of vehicle.




3. **Data preparation for modeling:**
   - We want to predict if the car crash was fatal or not.
   - Before we proceed to build a model, we'll have to encode categorical features.
   - We'll split the data into train and test to be able to evaluate the model that we build on the train data.

4. **Feature Engineering:** Here we create a new feature to find the age of the vehicle. We subtract year of vehicle bought from year of accident to know the same. We find below details:

```
Survived
Not_Survived    7.95508
survived        6.80422
```

**Mean/Average**

```
Survived
Not_Survived    8.00000
survived        6.00000
```

**Median/50% of data**

5. **Encoding:**
   - We encode the target variable by marking "Survived" as 1 and "Non-survived" as 0.
   - We then perform dummy encoding on all categorical variables, and we get data in this format now, making it much simpler for further analysis:

| dvcat_10-24 | dvcat_25-39 | dvcat_40-54 | dvcat_55+ | airbag_none | seatbelt_none | sex_m | abcat_nodeploy | abcat_unavail | occRole_pass |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 0 |

6. **Data Split:** Splitting the data in 70:30 ratio for train to test data:

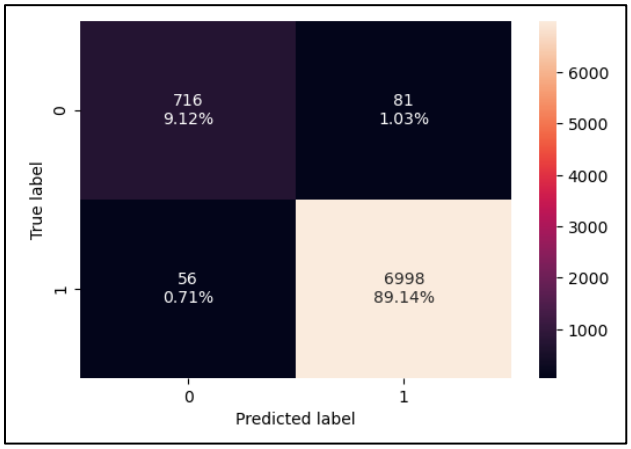
```
Shape of Training set :  (7851, 18)
Shape of test set :  (3366, 18)
Shape of Training set :  (7851,)
Shape of test set :  (3366,)
Percentage of classes in training set:
survived       0.89848
Not_Survived   0.10152
Name: Survived, dtype: float64
Percentage of classes in test set:
survived       0.88622
Not_Survived   0.11378
```

**<u>Topic 3: Model Building:</u> Part 1-Logistic Regression:** Is a statistical method used for binary classification tasks, where the goal is to predict the probability of a binary outcome (such as success/failure, yes/no, etc.) based on one or more predictor variables. Despite its name, logistic regression is actually a classification algorithm rather than a regression algorithm.

1. **Checking model performance on training set:**

| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| 0 | 0.98255 | 0.99206 | 0.98856 | 0.99031 |

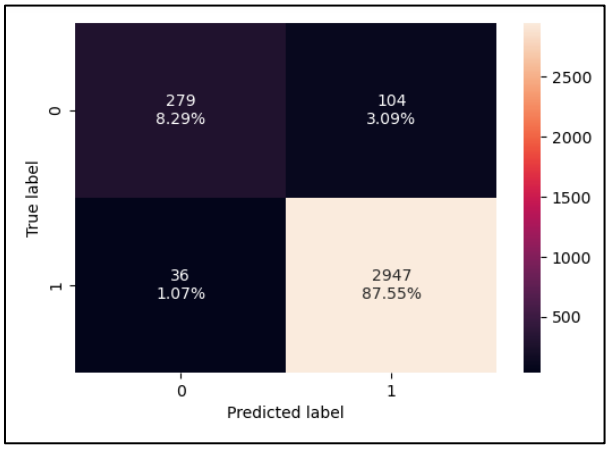
2. **Confusion Matrix on training set:**




3. **Checking model performance on test set:**

| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| 0 | 0.98039 | 0.99229 | 0.98568 | 0.98897 |

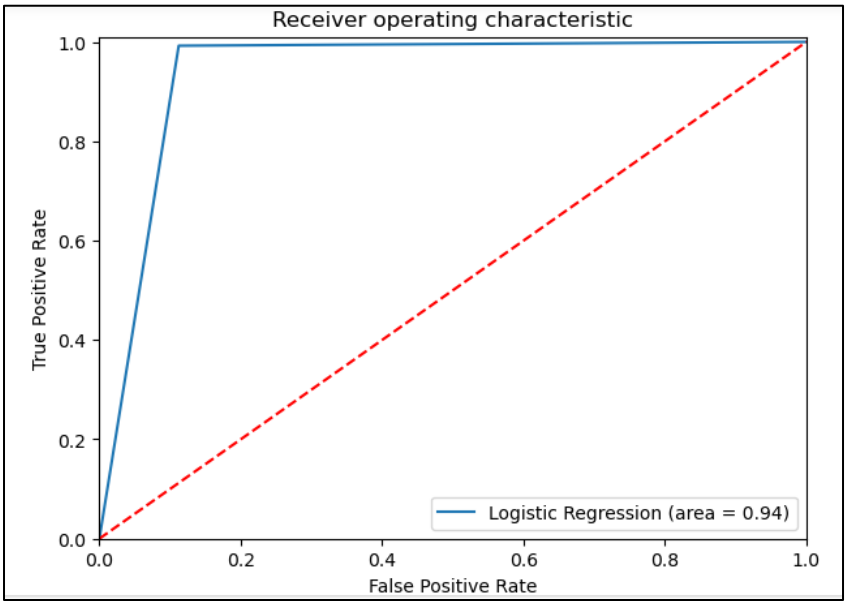
4. **Confusion Matrix on test set:**



5. **ROC-AUC:** When evaluating a Logistic Regression model, a higher ROC-AUC score indicates better discrimination between the positive and negative classes, suggesting that the model is better at ranking the instances correctly according to their predicted probabilities.

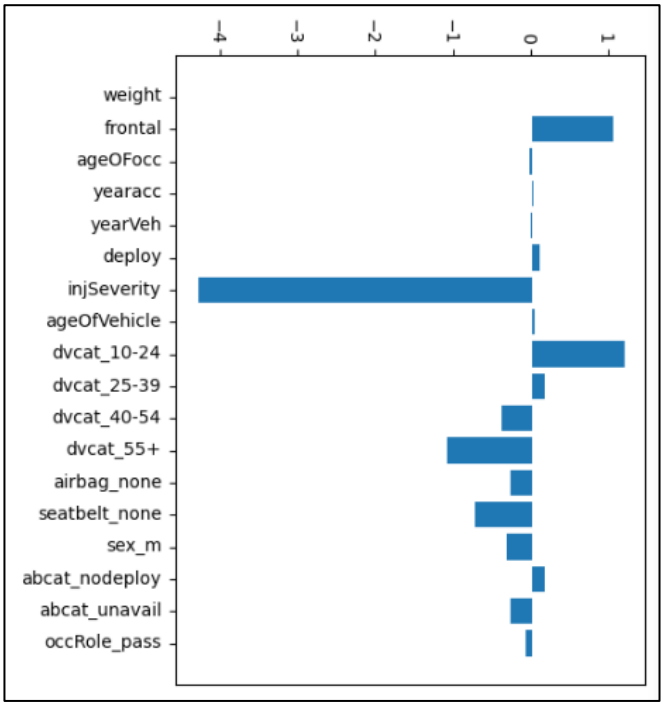
```
Train ROC-AUC score is : 0.9867268773352295

Test ROC-AUC score is : 0.9874860939580163
```




6. **Feature Importance:**

```
Feature: weight, Score: 0.00431
Feature: frontal, Score: 1.03968
Feature: ageOFocc, Score: -0.03866
Feature: yearacc, Score: 0.01859
Feature: yearVeh, Score: -0.01013
Feature: deploy, Score: 0.09951
Feature: injSeverity, Score: -4.29383
Feature: ageOfVehicle, Score: 0.02872
Feature: dvcat_10-24, Score: 1.18965
Feature: dvcat_25-39, Score: 0.16354
Feature: dvcat_40-54, Score: -0.38891
Feature: dvcat_55+, Score: -1.10087
Feature: airbag_none, Score: -0.27375
Feature: seatbelt_none, Score: -0.72715
Feature: sex_m, Score: -0.32717
Feature: abcat_nodeploy, Score: 0.16934
Feature: abcat_unavail, Score: -0.27375
Feature: occRole_pass, Score: -0.08803
```

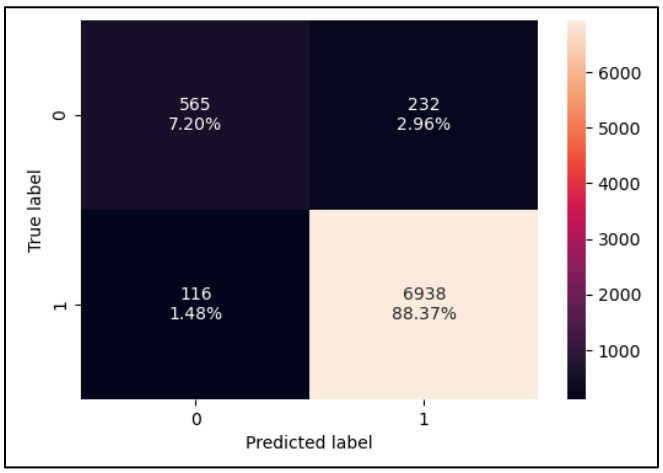


**Part 2: Building Linear Discrimant Analysis:** Linear Discriminant Analysis (LDA) is a classification algorithm used for both binary and multiclass classification tasks. It is a dimensionality reduction technique. It is used as a pre-processing step in Machine Learning and applications of pattern classification.

1. **Checking model performance on training set:**

| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| 0 | 0.95567 | 0.98356 | 0.96764 | 0.97553 |

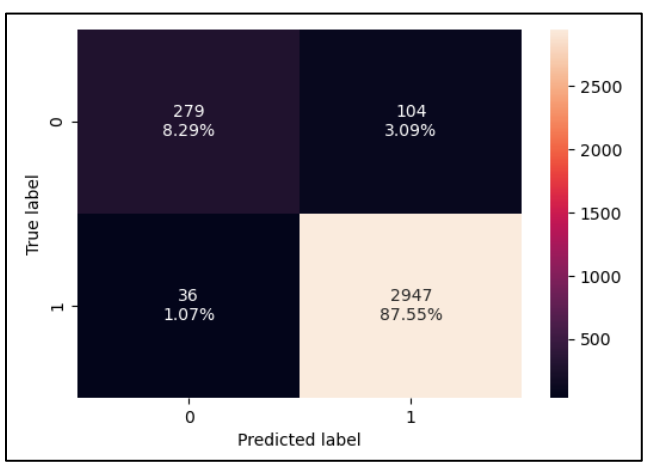
2. **Confusion Matrix on training set:**




3. **Checking model performance on test set:**

| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| 0 | 0.95841 | 0.98793 | 0.96591 | 0.97680 |

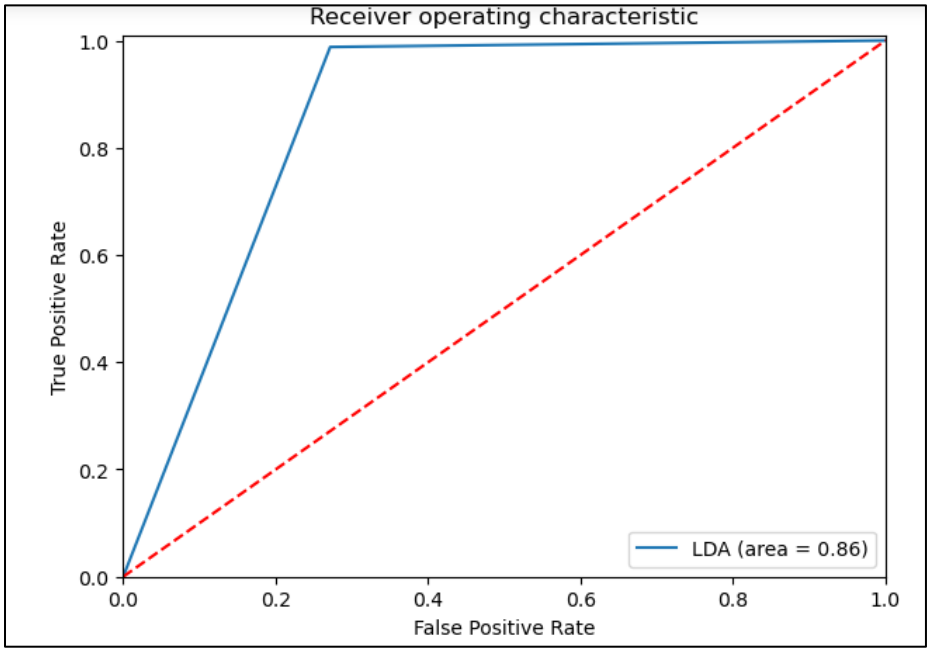
4. **Confusion Matrix on test set:**



5. **ROC-AUC:** ROC-AUC is particularly useful when the classes are imbalanced or when the cost of false positives and false negatives is different. It provides a single value that summarizes the model's performance across all possible classification thresholds.

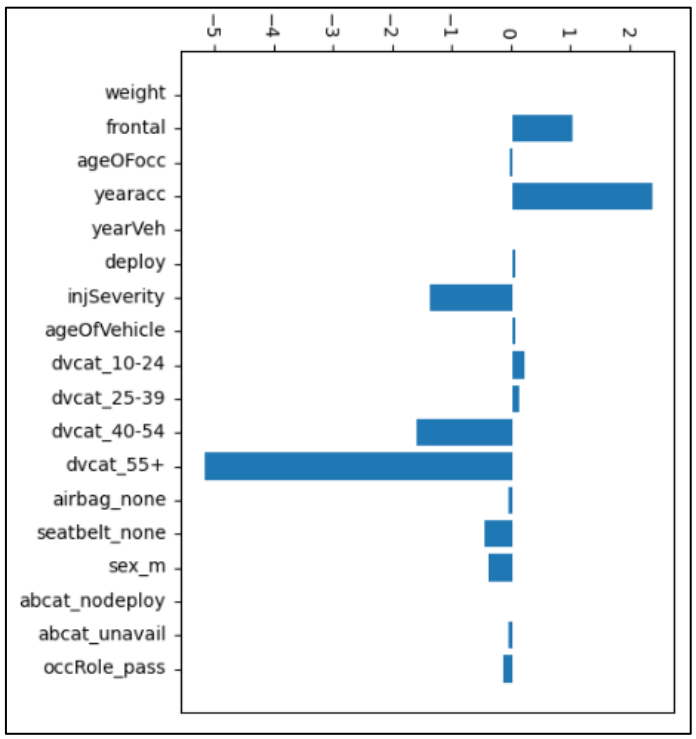
```
Train ROC-AUC score is : 0.9689817109027011

Test ROC-AUC score is : 0.9655270203914437
```




6. **Feature Importance:**

```
Feature: weight, Score: -0.00013
Feature: frontal, Score: 1.02274
Feature: ageOFocc, Score: -0.02970
Feature: yearacc, Score: 2.36381
Feature: yearVeh, Score: 0.01794
Feature: deploy, Score: 0.06156
Feature: injSeverity, Score: -1.38148
Feature: ageOfVehicle, Score: 0.04255
Feature: dvcat_10-24, Score: 0.19864
Feature: dvcat_25-39, Score: 0.12479
Feature: dvcat_40-54, Score: -1.59949
Feature: dvcat_55+, Score: -5.18418
Feature: airbag_none, Score: -0.06531
Feature: seatbelt_none, Score: -0.46753
Feature: sex_m, Score: -0.39722
Feature: abcat_nodeploy, Score: 0.00242
Feature: abcat_unavail, Score: -0.06531
Feature: occRole_pass, Score: -0.15153
```



## Topic 4: Model Performance Comparison and Final Model Selection:

1. **Training performance comparison:**

```
Training performance comparison:
```

| | Logistic Regression Base | Linear Discriminant Analysis Base |
|---|---|---|
| Accuracy | 0.98255 | 0.95567 |
| Recall | 0.99206 | 0.98356 |
| Precision | 0.98856 | 0.96764 |
| F1 | 0.99031 | 0.97553 |

2. **Test performance comparison:**

```
Test set performance comparison:
```

| | Logistic Regression Base | Linear Discriminant Analysis Base |
|---|---|---|
| Accuracy | 0.98039 | 0.95841 |
| Recall | 0.99229 | 0.98793 |
| Precision | 0.98568 | 0.96591 |
| F1 | 0.98897 | 0.97680 |

**Actionable Insights:**

- The model accuracy of logistic regression on both training data as well as testing data is almost same that is 98%.
- Similarly, AUC in logistic regression for training data and testing data is also similar.
- The other parameters of confusion matrix in logistic regression is also similar, therefore we can presume in this that our model is overfitted.
- We have therefore applied Grid Search CV to hyper tune our model and as per which F1 score in both training and test data was 98%.
- In case of LDA, the AUC for testing and training data is also same and it was 97%, besides this the other parameters of confusion matrix of LDA model was also similar and it clearly shows that model is overfitted here too.
- Score of Both Train and Test Data are quite near-by.

**Business Recommendations:**

- Overall we can conclude that logistic regression model is best suited for this data set given the level of accuracy in spite of the Linear Discriminant Analysis that the model is overfitted.